HYBRID PARTICIPATION INNOVATION LAB · STEP 1

An AI energy assistant for a solar energy community

Fair, shared solar energy for Rome's public housing, with rules written by the tenants

Next Wave

Mahsa Shahi · Kuhyar Saeedi · Farida Safdari · Gelila Tsehaye Melesse · Md Nazmul Islam · Pranto Argha Sarker

QUESTION 1 OF 6

What is the technology, and what problem does it solve?

THE PROBLEM

Families in public housing struggle to pay energy bills.

Solar panels on the roof help little: most energy is used in the evening, when there is no sun, and tenants have no say in how the money is shared.

OUR TECHNOLOGY: AN AI ENERGY ASSISTANT

1 **Predicts the sun:** how much energy the roof will make, hour by hour.

2 **Advises families:** the best time to run the washing machine or other appliances.

3 **Shares the money fairly,** using rules the tenants decide together.

QUESTION 2 OF 6

Who develops or promotes it?

A new idea

Nobody has built this yet. Our project designs it together with the people who will use it.

It builds on

Solar forecasting and smart plugs from energy and smart-home companies, and research from universities such as Tor Vergata.

Supporters and owners

The EU, the Italian state and GSE fund energy communities. The City of Rome supports them. ATER owns the buildings.

QUESTION 3 OF 6

Where will it be implemented?

A pilot in **one public housing complex** in Rome's eastern periphery, for example Tor Bella Monaca.

Large flat roofs

Plenty of space for solar panels that today produce nothing.

Families in need

Many low-income households who struggle with energy bills.

Near the university

Close to Tor Vergata, a possible research partner for on-site workshops.

QUESTION 4 OF 6

What benefits are expected?

Lower energy costs

More solar energy used at the right time means a bigger bonus for the community.

Fair sharing

The money is shared by rules the tenants themselves agree are fair.

Energy know-how

Families learn when and how to use energy better.

Cleaner energy

More local solar power, less CO2 and pollution.

QUESTION 5 OF 6

What uncertainties, risks or controversies may emerge?

Risk	In simple words
Fairness	People may disagree about what a "fair" share is.
Privacy	The AI needs private data about when people are at home.
Trust	Tenants may not trust a computer deciding about money.
Exclusion	Elderly people or people without smartphones may be left out.
Costs	It is not clear who pays for the devices and repairs.

QUESTION 6 OF 6

Why is participation necessary?

The AI decides who gets money, so the tenants must help write its rules.

What is "fair"?

Equal shares, more for poorer families, or more for those who shift their usage?

What data?

What the AI may know about their homes, and who can see it.

Advise or control?

Should it only suggest, or also switch appliances on?

STEP 1 · NEXT WAVE

An AI energy assistant for a solar energy community in Rome's public housing

The problem and the technology

Families in public housing struggle to pay energy bills. Our AI assistant predicts solar energy, tells families when to use appliances, and shares the money fairly.

Who develops or promotes it

A new idea. It builds on energy tech companies and university research. The state, GSE and the City of Rome support energy communities. ATER owns the buildings.

Where

A pilot in one public housing complex in Rome's periphery, for example Tor Bella Monaca, near Tor Vergata University.

Expected benefits

Lower energy costs. Money shared in a way tenants find fair. Families learn to use energy better. Cleaner energy, less pollution.

Risks and controversies

What is "fair"? Privacy of home data. Low trust in a computer deciding money. Elderly people left out. Who pays for devices?

Why participation is needed

The system decides how money is shared, so tenants must help write the rules. They know their needs best. Without them, nobody trusts or uses it.